Non-Essential Business-Cycles

Andreolli, Rickard & Surico (2025)

Lukas Boehnert

PhD elective, 2026/27

Week 1 reading group: a model presentation and discussion

How to use this deck

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⇒ Grey notes in the bottom left say what each slide is for

Part A: The paper

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- **Monetary transmission is mostly indirect** [Kaplan, Moll & Violante (2018)]
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$\Rightarrow$ If the rich cut restaurants and holidays, and the poor work in restaurants and hotels, the composition of demand becomes an amplification mechanism

Blueprint: start from a big, familiar idea, then narrow to one question

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- **Three settings**, three sources of variation:
  1. *Recessions:* average path from the NBER peak, 1973–2007 — time series
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- **Facts:**
  1. Non-essential *spending* is far more cyclical: -3.5% vs -1.5% a year into a recession
  2. Non-essential *earnings* even more so: $> -4\%$ vs insignificant after a 100bp hike
  3. Non-essential industries employ more *low-paid, hand-to-mouth* workers: 23% vs 11%

Blueprint: say where each fact comes from: data, variation, method

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  - $\rightarrow$ Alone (on top of TANK): modest, from 36% to 49% of the full response
- **Together:** the rich cut luxuries, the poor who make them lose income and cut everything
  - $\Rightarrow$ Consumption response to monetary policy $\sim 3\times$ TANK; a VAT on non-essentials only bites harder than a uniform VAT

Blueprint: say what each ingredient does to the result, not just that it is there

3. Contribution to the literature

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- **Heterogeneity and monetary transmission**
  - MPC–earnings covariance drives amplification [Bilbiie (2020), Patterson (2023)]
  - Indirect effects and earnings heterogeneity [Kaplan et al. (2018), Cloyne et al. (2020), Holm et al. (2021)]

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- **Demand composition over the cycle**
  - Durables [McKay & Wieland (2019), Beraja & Wolf (2021)]; trading down [Jaimovich et al. (2019)]; networks [Baqaee & Farhi (2018), Rubbo (2023)]

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- **Non-homothetic preferences**
  - Structural change [Boppart (2014), Comin et al. (2021)]; "luxuries are easier to postpone" [Browning & Crossley (2000)]

  ⇒ New: brought into an estimated business-cycle model

Blueprint: name 2–3 strands, one line each on what is new; not a list of 20 citations

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⇒ Details in Part B

Blueprint: three each, specific; not "identification could be better"

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   - Targeted VAT vs monetary policy

Empirical results

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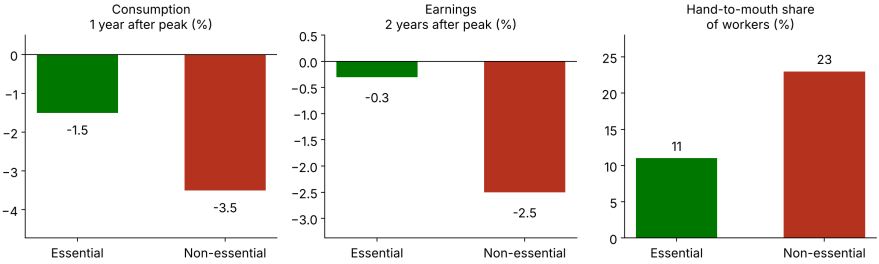
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Essential (elasticity < 1)	Non-essential (elasticity > 1)
Food at home (0.40), utilities (0.47)	Food away from home (1.37), new cars (1.74)
Rent (-1.1), health (0.81)	Owner-occupied housing (1.45), education (1.58)

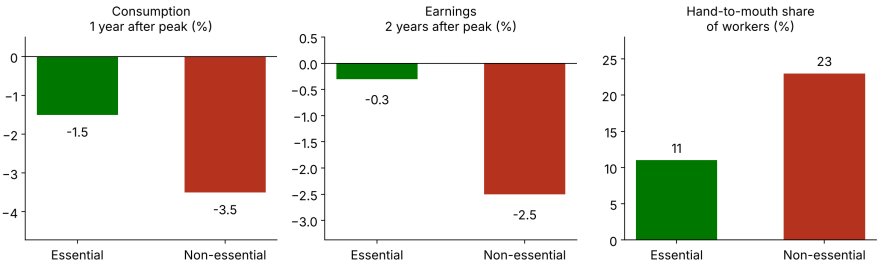
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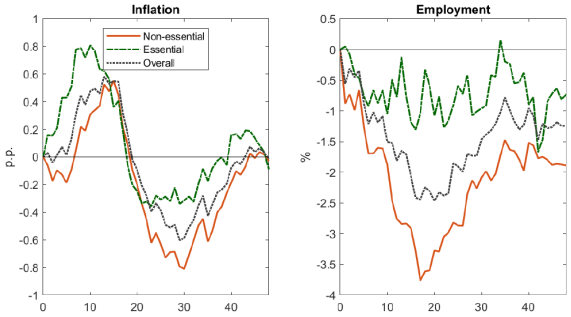


- Average US recession 1973–2007 (HP-detrended, from the NBER peak); HtM shares from the PSID
- Low earners are more likely in non-essentials (75% in the bottom decile, <55% at the top); median pay 12% lower

Bars drawn from numbers reported in Section 2.2 of the paper.

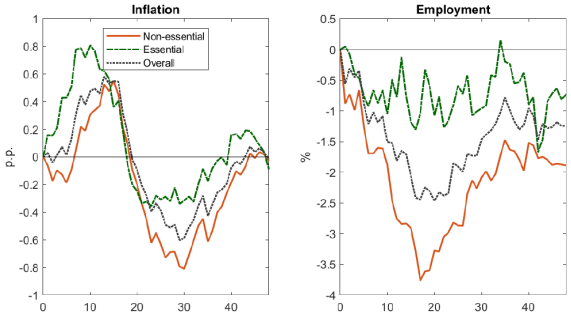
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⇒ But these are correlations: recessions have many causes

[Online Appendix, Figure F.5]

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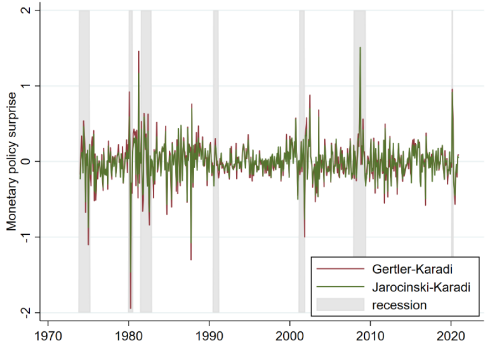
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  - Surprise as internal instrument (4 lags); extract the shock series
- **Instrument strength** [Montiel Olea & Pflueger (2013), critical value 12.04]
  - Gertler–Karadi surprise: $F = 13.9$ (baseline)
  - Jarociński–Karadi (info-robust): $F = 10.3$ (weak, appendix only)

Blueprint: state the identifying assumption in one sentence; link to what we did in class

The extracted shock series



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- Large spikes around 1980 and 2008: the extended series is model-generated before 1990

[Online Appendix, Figure G.1]

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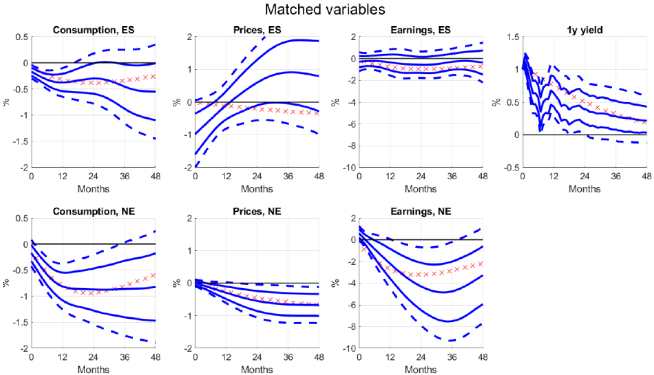
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Week 1 link: LP-IV = our L3 step; smoothing = shrinkage on the bias–variance trade-off [Montiel Olea et al. (2025)]

Blueprint: write the regression; say what is y, x, z, controls, horizon

Main result: non-essentials drive the aggregate response

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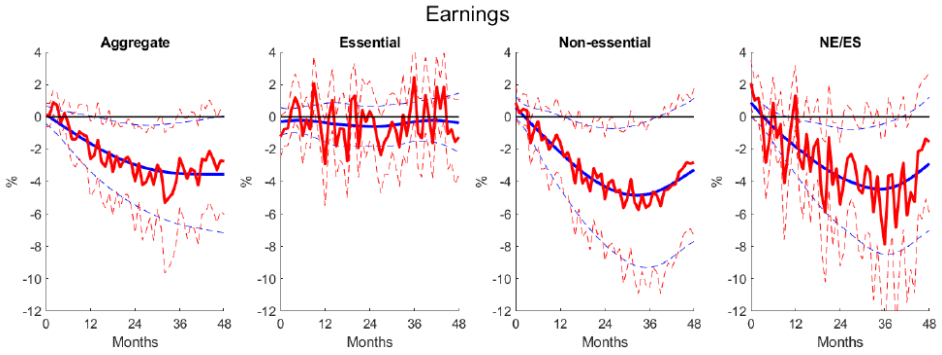
Blue: smooth LP-IV, 100bp rise in 1y yield (68/90% bands). Red $\times$: estimated model. Top: essentials; bottom: non-essentials.

$\Rightarrow$ Non-essential spending falls $\sim 2\times$ as much; non-essential earnings fall $>4\%$, essential $<1\%$

[Online Appendix, Figure J.1 (the matched panels of the paper's Figure 6)]

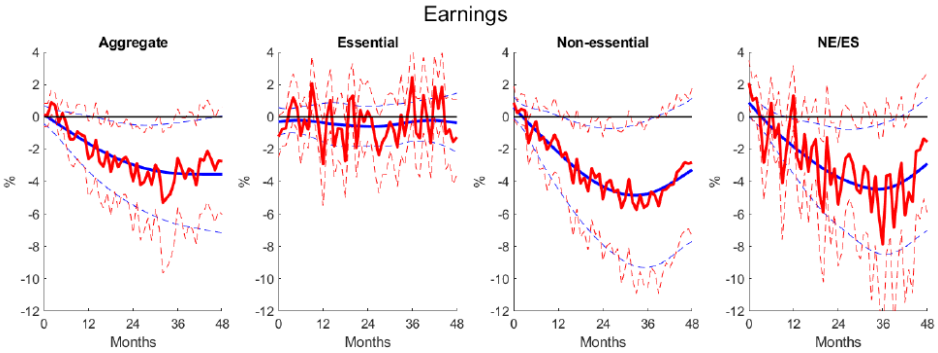
What smoothing does: SLP (blue) vs plain LP (red)

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- Earnings responses: same shape, but plain LP is very jagged for noisy monthly CPS data

What smoothing does: SLP (blue) vs plain LP (red)



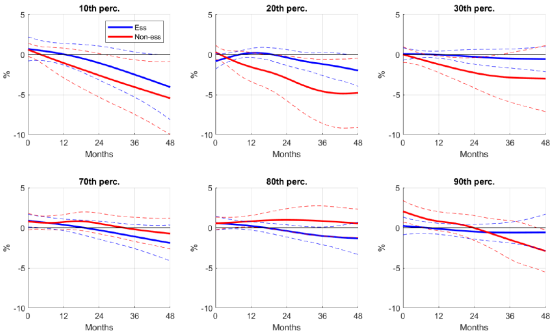
- Earnings responses: same shape, but plain LP is very jagged for noisy monthly CPS data
- Smoothing buys precision at the cost of some bias – the LP/VAR trade-off from today

[Online Appendix, Figure H.4 (earnings row)]

Blueprint: show one robustness figure that speaks to the method, list the rest

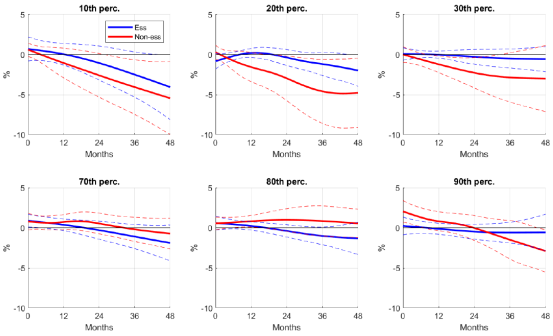
Along the earnings distribution

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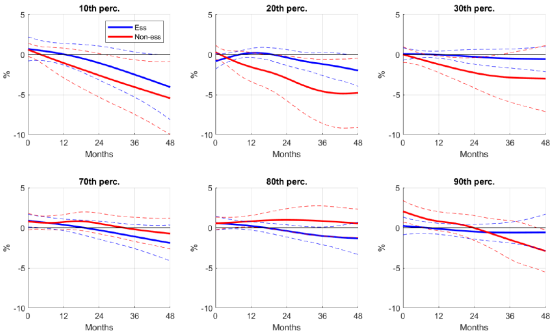
- Earnings percentiles 10–30 (top) and 70–90 (bottom); blue essential, red non-essential

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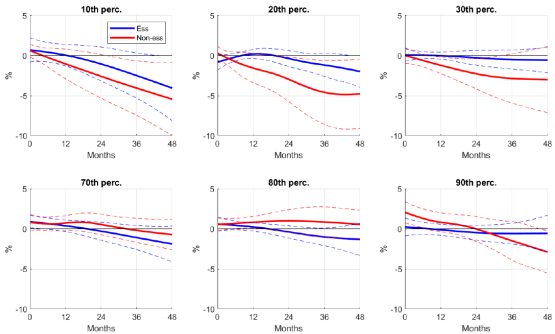
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Along the earnings distribution



[Online Appendix, Figure H.1]

- Earnings percentiles 10–30 (top) and 70–90 (bottom); blue essential, red non-essential
- **Within essentials:** little difference across deciles
- **Within non-essentials:** bottom deciles fall most
- Low earners in non-essentials lose 2–4$\times$ more than low earners in essentials

Robustness, in one slide

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- **Prices:** sectoral price responses insignificant $\Rightarrow$ the channel works through quantities and incomes, not relative prices

Model

Model: environment

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Blueprint: first tell the audience who lives in the economy, then their problems

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- **Ricardian** household H solves

$$\max_{\{C^E_{H,t}, C^N_{H,t}, B_{H,t}\}} \mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t \left[\frac{(C^E_{H,t})^{1-1/\gamma^E}}{1-1/\gamma^E} + \varphi \frac{(C^N_{H,t})^{1-1/\gamma^N}}{1-1/\gamma^N} - \xi \frac{N^{1+\chi}_{H,t}}{1+\chi} \right]$$

$$\text{s.t.} \quad P^E_t C^E_{H,t} + P^N_t C^N_{H,t} + B_{H,t} = W_{H,t} N_{H,t} + \Pi_{H,t} + T_{H,t} + R_{t-1} B_{H,t-1}$$

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$$\text{Engel curve } (k = H, L): \quad \varphi (C^N_{k,t})^{-1/\gamma^N} = \frac{P^N_t}{P^E_t} (C^E_{k,t})^{-1/\gamma^E}$$

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- Hours set by unions: $\xi N^{\chi}_{k,t} = w_{k,t} \zeta_{k,t}$; sticky information: plans updated with prob. λ

Households: homothetic preferences as a special case

Households: homothetic preferences as a special case

- Engel curve solved for non-essentials:

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$\Rightarrow$ The single parameter gap $\gamma^N - \gamma^E$ governs *both* the Engel curve (cross-section) and the different interest sensitivity of the two goods (time series) [Browning & Crossley (2000)]

Model: firms, market clearing and policy

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- **Firms**, in each sector $j \in \{E, N\}$: standard NK structure
  - CES final good producers and Calvo retailers $\Rightarrow$ a Phillips curve per sector
  - Competitive wholesalers combine low- and high-skill labour:

$$Y_t^j = A_t^j \left(N_{L,t}^j\right)^{\alpha^j} \left(N_{H,t}^j\right)^{1-\alpha^j}, \qquad \alpha^E < \alpha^N \quad \leftarrow \text{the only non-standard part}$$

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- CPI is a Fisher index (no ideal price index under non-homotheticity)

[Sections 4.3–4.4, Appendix D.3–D.4]

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  1. households and unions optimise, given prices and their (sticky) information
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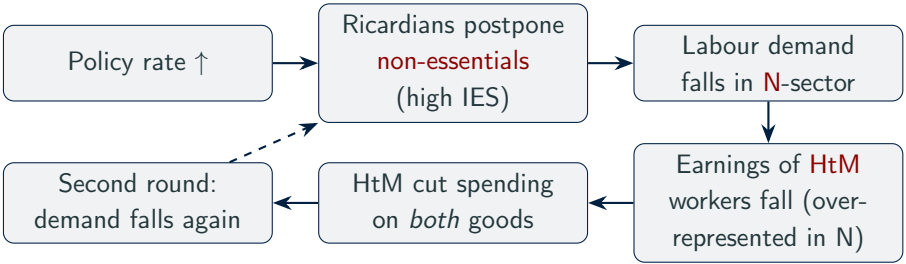
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- **Steady state:** zero inflation; φ and A^E/A^N set so essential budget shares are 0.44 (H) and 0.60 (L)
- **Solution:** log-linearise; truncate the sticky-information sums at 16 quarters [Verona & Wolters (2014)]

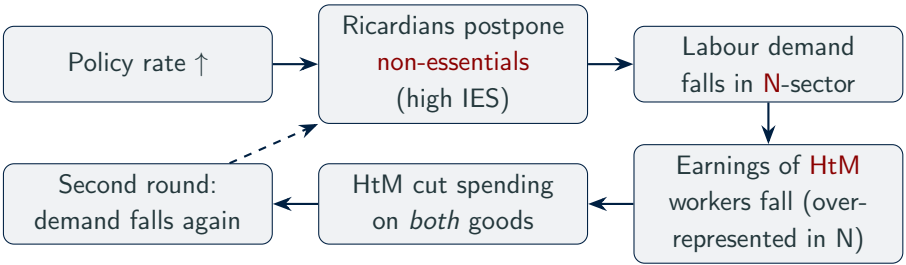
Blueprint: a clear equilibrium definition shows you understood the model

The mechanism

The mechanism

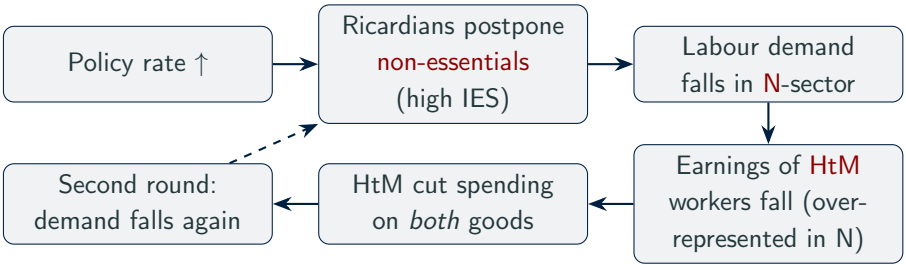


The mechanism



⇒ "The discretionary spending of the rich drives the earnings of the poor"

The mechanism



⇒ "The discretionary spending of the rich drives the earnings of the poor"

⇒ Needs all three: cyclical demand composition, uneven labour composition, high-MPC workers

Blueprint: a diagram beats three slides of algebra

Estimation: results

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- 7 parameters that govern the mechanism and the dynamics

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	Prior		Posterior
	Mean	SD	Mean
IES essentials γ^E	0.25	0.1	0.22
IES gap $\gamma^N - \gamma^E$	1.00	0.1	0.77
Low-skill share, E α^E	0.11	0.004	0.03
Low-skill gap $\alpha^N - \alpha^E$	0.12	0.004	0.32
Attentive share λ	0.90	0.03	0.01
Rate smoothing ρ_R	0.90	0.04	0.95
Calvo θ	0.90	1.00	0.96

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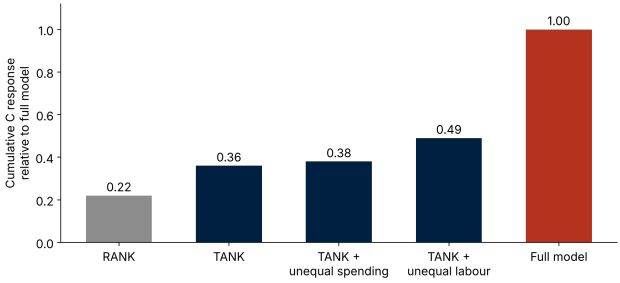
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Calibrated: $\beta = 0.99$, inverse Frisch 0.1, $\phi_\pi = 1.5$, $\phi_Y = 0.125$, $\mu^L = 1/3$; φ and A^E/A^N re-solved in each draw to hit the budget shares

[Table 1]

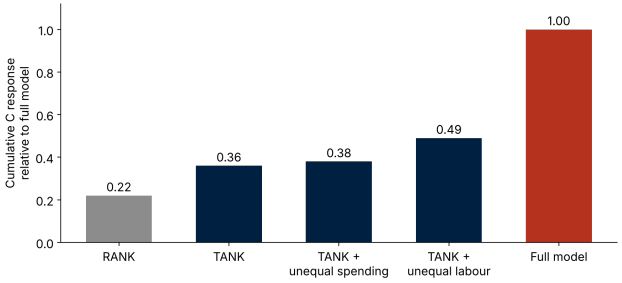
Where does amplification come from?

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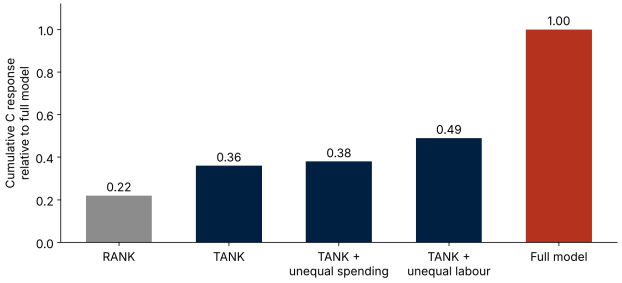
- Cumulative consumption response, relative to the full model

Where does amplification come from?



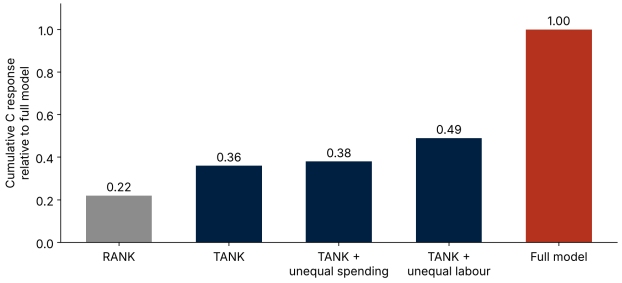
- Cumulative consumption response, relative to the full model
- RANK explains 22%, TANK 36%

Where does amplification come from?



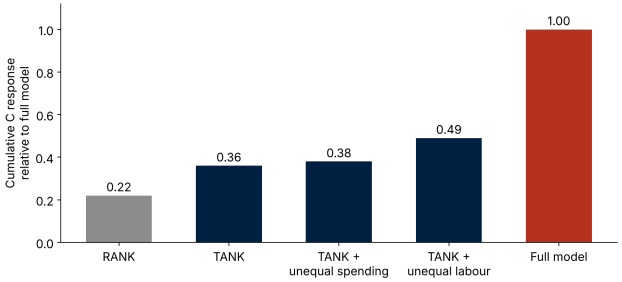
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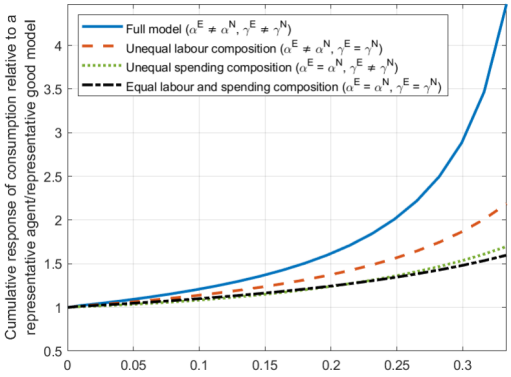
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$\Rightarrow$ A complementarity, not a sum

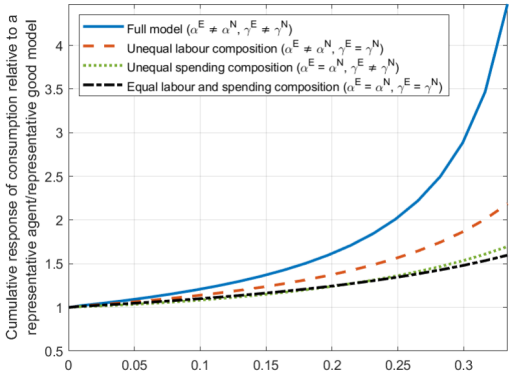
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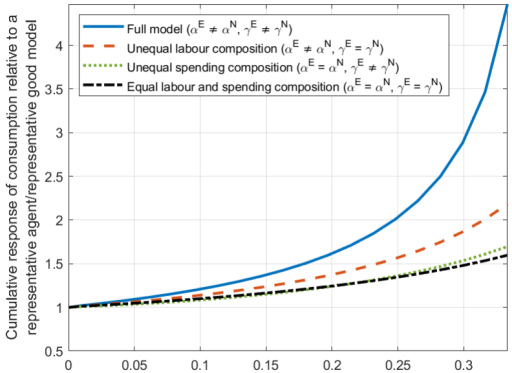
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[Online Appendix, Figure J.2; Section 6.2]

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- Full model: $\sim 4.5\times$ at $\mu^L = 1/3$
- **Analytical result:** with a representative agent, non-homotheticity does nothing for aggregates

  $\partial \hat{C}_t / \partial \varepsilon_t^{mp} = -\bar{\gamma}$ (average IES)

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$\Rightarrow$ A temporary VAT *cut* on non-essentials (e.g. UK "Eat Out to Help Out") is the strongest stimulus

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⇒ Fiscal policy can target it; monetary policy cannot

Blueprint: close Part A with the take-aways, then switch hats

Part B: Discussion

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$\Rightarrow$ Convincing on the mechanism; my comments are about measurement and how much the model can claim

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$\rightarrow$ Use a mechanical mapping: BEA PCE bridge tables (PCE $\rightarrow$ commodities) plus Census industry–NAICS crosswalks

$\rightarrow$ Show the continuous "downstream share" version (footnote 9 says it is robust)

$\rightarrow$ Placebo: reassign categories near elasticity 1 and drop housing; do the facts survive?

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- **Inference choices:** Newey–West with 12 lags of controls, plus smoothing with a CV penalty
  - Lag-augmented LPs need only robust SEs [Montiel Olea & Plagborg-Møller (2021)]

  → Report plain LP-IV with lag-augmented bands as a check on SLP coverage

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- **Which new fact does the model explain** that TANK cannot? [Karadi (2024)]

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$\rightarrow$ Panel LP-IV across euro area countries, interacted with non-essential employment shares

$\Rightarrow$ A natural class project

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  3. How would you identify the mechanism in Portugal, where tourism makes non-essentials tradable?

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Presenter (30 min)

- Question and context in 3 min
- Whole paper on one or two slides
- 2–3 literatures, what is new
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Discussant (10–15 min)

- Short summary (one slide)
- 2–3 main comments, ranked
- Each with a test or a fix
- Minor points, one line each
- One extension
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